

Ankur Desai

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EDUCATION

Michigan State University

B.S. in Computer Science

Expected, May 2028

East Lansing, MI

- **GPA:** 3.5 / 4.0
- **Relevant Coursework:** Linear Algebra, Discrete Mathematics, Object-Oriented Software Design, Computer Organization and Architecture, Data Structures & Algorithms

EXPERIENCE

Claude Builder Club @ MSU

Vice President & Co-founder

Oct. 2025 - Present

East Lansing, MI

- Architected full-stack member management platform deployed on CDN infrastructure with automated CI/CD pipeline using GitHub Actions, enforcing 85% code coverage thresholds, automated PostgreSQL schema migrations, and zero-downtime deployment strategy preventing production regressions.
- Engineered backend data layer with PostgreSQL implementing row-level security policies, atomic database functions for transactional event operations, and real-time data synchronization via logical replication streams, eliminating API polling overhead and maintaining sub-second UI update latency.
- Automated developer workflows with serverless integrations for member onboarding (workspace invites, GitHub team synchronization, Slack notifications), and standardized code review processes, reducing new contributor setup from 2+ hours to 5 minutes.

MindMosaic — Relationship Powered Note Taking Tool

Product Manager

Aug. 2025 – Nov. 2025

East Lansing, MI

- Directed a team of 4 to build a full-stack note-taking app with a chapter-and-page model, layered Markdown editing, and visual concept mapping via annotated links.
- Integrated React (Tailwind, marked, OAuth2), and established CI/CD workflows to reach 94% code coverage.
- Designed PostgreSQL + Neo4j stack, boosting query speed by 48% and reducing server load 30%.

PROJECTS

Mira — Distributed Intelligence Tool | *FastAPI, Next.js, AWS (EC2, RDS)* May 2025 – Nov. 2025

End-to-end voice intelligence system, aggregating real-time audio streams to execute context-aware automation

- Architected a thread-safe multi-tenant service registry using LRU eviction and TTL-based resource cleanup to manage lifecycle for 100+ concurrent networks, preventing memory leaks in long-running server instances.
- Engineered a real-time Digital Signal Processing (DSP) engine utilizing Welch's PSD and windowed variance analysis to score audio quality across 10+ concurrent streams with <100ms latency.
- Built a production hybrid search pipeline merging BM25 keyword matching with Sentence-BERT embeddings (384-dim), implementing 5-minute conversation boundary detection to optimize context retrieval accuracy.
- Deployed containerized FastAPI microservice via GitHub Actions CI/CD pipeline to AWS EC2, implementing automated Alembic migrations and RDS connection pooling (5 core + 10 overflow) for zero-downtime updates.

WizViz — SpartaHack X | *Python, MediaPipe, OpenCV, NumPy*

Feb. 2025

Head-to-head combat game utilizing pose estimation and gesture recognition **Winner:** Computer Vision Track

- Engineered a low-latency gesture pipeline using MediaPipe Pose to map 33-point skeletal landmarks to game actions, achieving 60 FPS with sub-20ms input lag by optimizing the inference loop.
- Developed a custom vector-based physics engine handling AABB collision detection to map 3D real-world movements into 2D game state in real-time.

TECHNICAL SKILLS

Languages Python, C++, Java, JavaScript/TypeScript, SQL, Bash

AI & Computer Vision PyTorch, TensorFlow, MediaPipe, Pyannote.audio, Sentence-Transformers

Full-Stack Development React, Next.js, FastAPI, Electron, SQLAlchemy, Node.js

DevOps & Cloud AWS (Lambda, EC2, RDS), Docker, GitHub Actions, Linux, PostgreSQL